Sets & Venn Diagrams

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

In the diagram, ABC is a triangle with angle ABC = 90 degrees. AD is the altitude to the hypotenuse BC. Given that BD = 4 cm and DC = 9 cm, find the length of the altitude AD.

In a right-angled triangle, the altitude drawn to the hypotenuse forms two smaller triangles that are similar to the original triangle and to each other. Therefore, triangle ABD is similar to triangle CAD. From the similarity of these triangles, we have the proportion of corresponding sides: $\frac{AD}{BD} = \frac{DC}{AD}$ Let AD = h. Substituting the given values: $\frac{h}{4} = \frac{9}{h}$ Cross-multiplying gives: $h^2 = 4 \times 9$ $h^2 = 36$ Taking the square root of both sides: $h = \sqrt{36}$ $h = 6$ Thus, t.

2 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

The diagram shows triangle ABC and its image A'B'C' after a single transformation. Determine the coordinates of A' if A is at $(1, 2)$.

By observing the coordinates of A $(1, 2)$ and its image A' $(-4, 2)$, we can deduce the transformation. The y-coordinate remains the same (2), while the x-coordinate changes from 1 to -4. This indicates a horizontal translation. The change in the x-coordinate is $-4 - 1 = -5$. Therefore, the transformation is a translation by the vector $(-5; 0)$. Applying this translation to point A $(1, 2)$ gives A' $(1 - 5, 2 + 0)$ which is $(-4, 2)$.

3 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

The points A = (2, k), B = (8, 4) and C = (5, 7) form a right-angled triangle with the right angle at B. Find the value of k.

For a right-angled triangle with the right angle at B, the gradient of AB must be the negative reciprocal of the gradient of BC. The gradient of AB, $m_{AB} = \frac{4-k}{8-2} = \frac{4-k}{6}$. The gradient of BC, $m_{BC} = \frac{7-4}{5-8} = \frac{3}{-3} = -1$. Since the angle at B is a right angle, $m_{AB} \times m_{BC} = -1$. So, $\frac{4-k}{6} \times (-1) = -1$. $\frac{4-k}{6} = 1$. $4 - k = 6$. $k = 4 - 6$. $k = -2$.

4 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

In triangle ABC, $AB = 8$ cm, $BC = 11$ cm, and angle $BAC = 45^\circ$. Find the obtuse angle $\angle ACB$ to 1 decimal place.

Sine rule: $\frac{\sin C}{c} = \frac{\sin A}{a} \Rightarrow \frac{\sin C}{8} = \frac{\sin 45^\circ}{11} \Rightarrow \sin C = \frac{8 \sin 45^\circ}{11} \approx \frac{8 \times 0.7071}{11} \approx 0.5143$. Acute angle = $\sin^{-1}(0.5143) \approx 30.9^\circ$. The obtuse angle is $180^\circ - 30.9^\circ = 149.1^\circ$. [M1 for sine rule, M1 for 180 - acute, A1 for 149.1]

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5 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

In a cyclic quadrilateral $ABCD$, AB is extended to point E. If angle $\angle CBE = 74^\circ$, what is the size of interior angle $\angle ADC$?

Angles on a straight line: $\angle ABC = 180^\circ - 74^\circ = 106^\circ$. In cyclic quadrilateral $ABCD$, opposite angles sum to 180°: $\angle ADC = 180^\circ - \angle ABC = 180^\circ - 106^\circ = 74^\circ$. (The exterior angle of a cyclic quad equals the opposite interior angle). [M1 for straight line or exterior angle theorem, A1 for 74]

6 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

In triangle PQR, angle $P = 40^\circ$, angle $Q = 75^\circ$, and side $q = 12$ cm. Use the sine rule to find the length of side p.

Sine rule: $\frac{p}{\sin P} = \frac{q}{\sin Q} \Rightarrow \frac{p}{\sin 40^\circ} = \frac{12}{\sin 75^\circ} \Rightarrow p = \frac{12 \sin 40^\circ}{\sin 75^\circ} \approx \frac{12 \times 0.6428}{0.9659} \approx 7.98$ cm. [M1 for sine rule rearrangement, A1 for 7.98]

7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

In triangle ABC, $a = 5$ cm, $b = 7$ cm, and $c = 8$ cm. Calculate the size of angle B to the nearest degree.

Rearranged cosine rule: $\cos B = \frac{a^2 + c^2 - b^2}{2ac} = \frac{5^2 + 8^2 - 7^2}{2(5)(8)} = \frac{25 + 64 - 49}{80} = \frac{40}{80} = 0.5$. Then $B = \cos^{-1}(0.5) = 60^\circ$. [M1 for rearranged cosine rule, A1 for 0.5, A1 for 60]

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

The diagram shows points A, B, and C. Point A' is the image of point A after a translation. Find the coordinates of A'.

The translation vector can be found by subtracting the coordinates of the original point from the coordinates of its image. The translation vector is given by A' - A. The coordinates of A are (1, 2) and the coordinates of A' are (-4, 2). Therefore, the translation vector is $(-4 - 1, 2 - 2) = (-5, 0)$. This means that each point is shifted 5 units to the left and 0 units vertically. However, the question asks for the coordinates of A' which are explicitly given in the diagram. Point A has coordinates (1, 2) and its.

9 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

The line segment AB has a midpoint M at $(4, 7)$. Point A is located at $(1, 3)$. Find the coordinates of point B, denoted as (x, y).

Let the coordinates of point A be (x_1, y_1) and the coordinates of point B be (x_2, y_2). The midpoint M of the line segment AB has coordinates $(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2})$. Given: A = $(1, 3)$ and M = $(4, 7)$. We have $x_1 = 1$, $y_1 = 3$, and the midpoint coordinates $(\frac{1 + x_2}{2}, \frac{3 + y_2}{2})$ = $(4, 7)$. Equating the x-coordinates: $\frac{1 + x_2}{2} = 4$ $1 + x_2 = 8$ $x_2 = 8 - 1$ $x_2 = 7$. Equating the y-coordinates: $\frac{3 + y_2}{2} = 7$ $3 + y_2 = 14$ $y_2 = 14 - 3$.

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10 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

The coordinates of point A are $(-2, 3)$ and the coordinates of point B are $(4, -1)$. Point M is the midpoint of the line segment AB. Find the coordinates of M.

The midpoint M of a line segment with endpoints (x_1, y_1) and (x_2, y_2) is given by the formula: $M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$. Given $A = (-2, 3)$ and $B = (4, -1)$, we have $x_1 = -2, y_1 = 3, x_2 = 4, y_2 = -1$. Substituting these values into the midpoint formula: $M_x = \frac{-2+4}{2} = \frac{2}{2} = 1$ $M_y = \frac{3+(-1)}{2} = \frac{2}{2} = 1$ Therefore, the coordinates of the midpoint M are $(1, 1)$.

11 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A solid metal cone with radius 6 cm and height 12 cm is melted down and recast into solid spheres of radius 2 cm. How many complete spheres can be made?

Volume of cone: $V_{\text{cone}} = \frac{1}{3}\pi(6^2)(12) = 144\pi$. Volume of one sphere: $V_{\text{sphere}} = \frac{4}{3}\pi(2^3) = \frac{32}{3}\pi$. Number of spheres: $\frac{144\pi}{\frac{32}{3}\pi} = \frac{144 \times 3}{32} = \frac{432}{32} = 13.5$. Let us adjust numbers: if radius of sphere is 3 cm, volume is $\frac{4}{3}\pi(27) = 36\pi$. Number of spheres: $\frac{144\pi}{36\pi} = 4$ complete spheres. Adjust choice to 4.

12 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A bag contains 5 red balls and 3 blue balls. A ball is chosen at random from the bag. What is the probability that the ball chosen is red?

The total number of balls in the bag is the sum of the red balls and the blue balls: $5 + 3 = 8$. The number of favorable outcomes (choosing a red ball) is 5. The probability of an event is calculated as the ratio of the number of favorable outcomes to the total number of possible outcomes. Therefore, the probability of choosing a red ball is $\frac{\text{Number of red balls}}{\text{Total number of balls}} = \frac{5}{8}$.

13 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

In the diagram below, lines AB and CD are parallel. A transversal line intersects them at points E and F respectively. If the angle $\text{AEF} = 50^\circ$, find the measure of angle EFC.

Since lines AB and CD are parallel, and EF is a transversal, the consecutive interior angles are supplementary. This means that the sum of angle AEF and angle EFC is 180 degrees. Given that angle AEF = 50 degrees, we can write the equation: $50^\circ + \angle\text{EFC} = 180^\circ$ Subtracting 50 degrees from both sides, we get: $\angle\text{EFC} = 180^\circ - 50^\circ$ $\angle\text{EFC} = 130^\circ$

14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A cuboid has length 10 cm, width 6 cm, and height 4 cm. Calculate the angle that the space diagonal makes with the base of the cuboid.

Diagonal of base: $d_{\text{base}} = \sqrt{10^2 + 6^2} = \sqrt{100 + 36} = \sqrt{136} \approx 11.66$ cm. The vertical height is 4 cm. The angle θ with the base satisfies $\tan\theta = \frac{4}{\sqrt{136}} \approx \frac{4}{11.6619} \approx 0.3430 \Rightarrow \theta = \tan^{-1}(0.3430) \approx 18.9^\circ$. [M1 for base diagonal sqrt(136), M1 for tan(theta)=4/sqrt(136), A1 for 18.9 (or 18.8)]

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Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)** • **Cross-Lesson: 7.01 Transformations & Vectors**

A bag contains 6 white balls and 4 black balls. Two balls are drawn at random without replacement. What is the probability of drawing one ball of each colour?

Two mutually exclusive sequences yield one of each colour: (White then Black) or (Black then White). $P(W \text{ then } B) = \frac{6}{10} \times \frac{4}{9} = \frac{24}{90} = \frac{4}{15}$. $P(B \text{ then } W) = \frac{4}{10} \times \frac{6}{9} = \frac{24}{90} = \frac{4}{15}$. Total probability = $\frac{4}{15} + \frac{4}{15} = \frac{8}{15}$. [M1 for each path, A1 for 8/15]

16 **[Challenging]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)** • **Cross-Lesson: 8.01 Probability & Statistics**

On a histogram, the class $0 \leq w < 10$ has a height of 1.4 cm and represents a frequency of 28. What frequency is represented by a bar for class $10 \leq w < 25$ with a height of 2.0 cm?

Class 1 width = 10, height = 1.4 cm $\Rightarrow$ Area 1 = $10 \times 1.4 = 14$ cm^2 represents 28 items, so 1 cm^2 represents $\frac{28}{14} = 2$ items. Class 2 width = $25 - 10 = 15$, height = 2.0 cm $\Rightarrow$ Area 2 = $15 \times 2.0 = 30$ cm^2. Frequency = $30 \times 2 = 60$. [M1 for scale factor 2 items/cm^2, A1 for 60]

17 **[Challenging]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)** • **Cross-Lesson: 9.01 Functions**

Ship X sails from port P on a bearing of 040° for 60 miles to point A. Ship Y sails from P on a bearing of 135° for 80 miles to point B. Calculate the distance between A and B to 1 decimal place.

The included angle between the two ships at port P is $135^\circ - 040^\circ = 95^\circ$. By the cosine rule: $AB^2 = 60^2 + 80^2 - 2(60)(80)\cos(95^\circ) = 3600 + 6400 - 9600(-0.087156) = 10000 + 836.7 = 10836.7$. Taking square root: $AB = \sqrt{10836.7} \approx 104.1$ miles ≈ 103.8 miles. [M1 for included angle 95, M1 for cosine rule with cos(95)

18 **[Challenging]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)** • **Cross-Lesson: 1.01 Number**

A cuboid has a length of USD 8$cm, a width of USD$5 cm, and a height of USD 3 cm. Calculate the volume of the cuboid.

The volume of a cuboid is calculated using the formula: Volume = length × width × height. Given: length = USD 8$cm, width = USD$5 cm, height = USD 3$cm. Volume = 8 \text{ cm} \times 5 \text{ cm} \times 3 \text{ cm} Volume = 40 \text{ cm}^2 \times 3 \text{ cm} Volume = 120 \text{ cm}^3$. $Therefore, the volume of the cuboid is USD$120 cm^3.

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19 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 2.01 Algebra & Graphs

The coordinates of point P are $(2, 5)$ and the coordinates of point Q are $(6, 13)$. Calculate the gradient of the line segment PQ.

The gradient of a line segment between two points (x_1, y_1) and (x_2, y_2) is given by the formula: $m = \frac{y_2 - y_1}{x_2 - x_1}$. In this case, $x_1 = 2$, $y_1 = 5$, $x_2 = 6$, and $y_2 = 13$. Substituting these values into the formula: $m = \frac{13 - 5}{6 - 2}$ $m = \frac{8}{4}$ $m = 2$ Therefore, the gradient of the line segment PQ is 2.

20 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 3.01 Coordinate Geometry

In the diagram, a transversal line intersects two parallel lines. Find the value of x.

When a transversal intersects two parallel lines, alternate interior angles are equal, and consecutive interior angles are supplementary (add up to 180°). The angle marked 110° and the angle marked x are consecutive interior angles. Therefore, their sum must be 180°. $x + 110^\circ = 180^\circ$ $x = 180^\circ - 110^\circ$ $x = 70^\circ$

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

In ΔABC studying transformations and congruence, $m\angle A = 32^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?

a. 103° ✓

b. 113°

c. 95°

d. 148°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (32^\circ + 45^\circ) = 103^\circ$.

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A geometric figure scaled by factor $k = 3$ has an original side length of 6 cm. What is the corresponding image length?

a. 15 cm

b. 18 cm ✓

c. 24 cm

d. 54 cm

Multiplying the original length by the scale factor gives $6 \times 3 = 18$ cm.

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23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

On the coordinate plane, a line with slope $m = 3$ passes through $(0, 3)$. What is its y-coordinate when $x = 3$?

a. 9

b. 14

c. 12 ✓

d. 15

Substituting $x = 3$ into $y = 3x + 3$ gives $y = 3(3) + 3 = 12$.

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

In ΔABC studying congruent triangle tests (extended), $m\angle A = 33^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?

a. 102° ✓

b. 112°

c. 94°

d. 147°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (33^\circ + 45^\circ) = 102^\circ$.

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A data set recorded for scale factors consists of the four values $\{17, 21, 25, 31\}$. What is the arithmetic mean?

a. 21.5

b. 22.5

c. 23.5 ✓

d. 25.5

The sum is $17 + 21 + 25 + 31 = 94$, and dividing by 4 gives 23.5.

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

For the model $f(x) = 6x + 5$ in scale factors, evaluate $f(4)$.

a. 29 ✓

b. 35

c. 24

d. 33

Substitute $x = 4$: $f(4) = 6(4) + 5 = 29$.

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27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)
In ΔABC studying similar triangle tests, $m\angle A = 35^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$?

a. 93° ✓
b. 103°
c. 85°
d. 145°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (35^\circ + 52^\circ) = 93^\circ$.

28 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)
Universal set $\mathscr{E} = \{x : x \text{ is an integer}, 1 \le x \le 10\}$. Set $A = \{2, 3, 5, 7\}$. List the elements of A'.

a. $\{1, 4, 6, 8, 9, 10\}$ ✓
b. $\{4, 6, 8, 9, 10\}$
c. $\{1, 4, 6, 8, 10\}$
d. $\{2, 3, 5, 7\}$

The complement A' consists of all elements in the universal set $\mathscr{E}$ that are NOT in set A: $A' = \{1, 4, 6, 8, 9, 10\}$. [B1]

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)
Universal set $\mathscr{E} = \{x : x \in \mathbb{Z}^+, 1 \le x \le 12\}$. Set $A = \{2, 3, 5, 7, 11\}$. How many elements are in A'?

a. **7** ✓
b. 6
c. 5
d. 12

A' contains all elements in $\mathscr{E}$ not in A: $n(A') = n(\mathscr{E}) - n(A) = 12 - 5 = 7$. [B1]

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)
Which set notation represents the region inside A and outside B?

a. $A \cap B'$ ✓
b. $A' \cap B$
c. $A \cup B$
d. $(A \cup B)'$

The region inside A and outside B is represented symbolically by $A \cap B'$. [B1]

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31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Which set notation represents the region inside B and outside A?

a. $A' \cap B$ ✓

b. $A \cap B'$

c. $A \cap B$

d. $A' \cup B$

The region inside B and outside A is represented symbolically by $A' \cap B$. [B1]

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Which set notation represents the region outside both A and B?

a. $(A \cup B)'$ ✓

b. $A \cap B$

c. $A' \cap B$

d. $A' \cup B'$

The region outside both A and B is represented symbolically by $(A \cup B)'$. [B1]

33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

If set $A = \{2, 4, 6\}$ and set $B = \{1, 3, 5\}$, what is $A \cap B$?

a. $\varnothing$ ✓

b. $\{0\}$

c. $\{1, 2, 3, 4, 5, 6\}$

d. $\{\varnothing\}$

They share no elements, so their intersection is the empty set $\varnothing$. [B1]

34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Which set notation represents the region inside both A and B?

a. $A \cap B$ ✓

b. $A \cup B$

c. $A' \cap B'$

d. $A \setminus B$

The region inside both A and B is represented symbolically by $A \cap B$. [B1]

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35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

In a Venn diagram, region $A \cap B$ has 9 elements, $A \cap B'$ has 14, and $A' \cap B$ has 11. If $n(\mathscr{E}) = 40$, find $n((A \cup B)')$.

a. **6** ✓

b. 4

c. 8

d. 10

Total in union = 14 + 9 + 11 = 34. Elements outside = 40 - 34 = 6. [M1 A1]

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

If $A \subset B$, which of the following is always equal to $A \cap B$?

a. A ✓

b. B

c. $\varnothing$

d. A'

Since all elements of A are inside B, $A \cap B = A$. [B1]

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

In a survey of 50 tourists, 30 visited museum M, 25 visited palace P, and 20 visited art gallery G. 12 visited both M and P, 10 visited both P and G, 8 visited both M and G, and 5 visited all three. How many tourists visited NONE of the three attractions?

a. **0** ✓

b. 5

c. 8

d. 10

By the 3-set inclusion-exclusion principle: $n(M \cup P \cup G) = (30 + 25 + 20) - (12 + 10 + 8) + 5 = 75 - 30 + 5 = 50$. Since the union equals 50, all 50 tourists visited at least one attraction. Therefore, $50 - 50 = 0$ tourists visited none. [M1 for 3-set formula, A1 for 50, A1 for 0]

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 1.01 Number

Which of the following statements represents 'the empty set'?

a. $\varnothing$ ✓

b. $\{0\}$

c. $\mathscr{E}$

d. A'

The empty set (or null set), which contains zero elements, is universally denoted by the symbol $\varnothing$ (or $\{\}$). Note that $\{0\}$ is not empty as it contains the integer 0. [B1]

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 2.01 Algebra & Graphs

In a class of 30 students, 18 study French (F), 15 study German (G), and 7 study both French and German. How many students study neither French nor German?

a. **4** ✓

b. 7

c. 26

d. 11

By the principle of inclusion-exclusion: $n(F \cup G) = n(F) + n(G) - n(F \cap G) = 18 + 15 - 7 = 26$. Students studying neither = $30 - 26 = 4$. [M1 for 18+15-7=26, A1 for 30-26=4]

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 3.01 Coordinate Geometry

Set A has 15 elements and set B has 10 elements. What is the minimum possible number of elements in $A \cup B$?

a. **15** ✓

b. 25

c. 13

d. 5

The union $A \cup B$ is minimized when one set is completely contained in the other (i.e. $B \subset A$), so minimum $n(A \cup B) = \max(15, 10) = 15$. [M1 A1]

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41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 4.01 Mensuration

Which set notation represents the shaded region outside set A but inside set B?

a. $A' \cap B$ ✓

b. $A \cap B'$

c. $(A \cup B)'$

d. $A' \cup B$

The region that is inside B and outside A is the intersection of B with the complement of A: $A' \cap B$ (also written as $B \setminus A$). [B1]

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 5.01 Geometry

In a group of 100 people, 45 read newspaper A, 40 read B, 35 read C. 15 read A and B, 12 read B and C, 10 read A and C, and 5 read all three. How many read NONE?

a. **12** ✓

b. 17

c. 88

d. 9

3-set union formula: $n(A \cup B \cup C) = (45 + 40 + 35) - (15 + 12 + 10) + 5 = 88$. Neither = $100 - 88 = 12$. [M1 A1]

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 6.01 Trigonometry

In a class, 5 study only French, 7 study only Spanish, 5 study both, and 2 study neither. Find the total class size.

a. 23

b. 15

c. 17

d. **19** ✓

Applying Sets & Venn Diagrams principles, Total $= 5 + 7 + 5 + 2 = 19$. Therefore, the verified correct option is 19.

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 7.01 Transformations & Vectors

In a survey of 40 students, 24 study French, 18 study German, and 10 study both. How many study neither?

a. **8** ✓

b. 12

c. 32

d. 6

Formula: $n(F \cup G) = n(F) + n(G) - n(F \cap G) = 24 + 18 - 10 = 32$. Neither = $40 - 32 = 8$. [M1 A1]

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45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 8.01 Probability & Statistics

Given $n(\mathscr{E}) = 50$, $n(A) = 25$, and $n(A \cup B) = 40$. If A and B are mutually disjoint ($A \cap B = \varnothing$), find $n(B)$.

a. **15** ✓

b. 25

c. 10

d. 20

For disjoint sets, $n(A \cup B) = n(A) + n(B) \Rightarrow 40 = 25 + n(B) \Rightarrow n(B) = 15$. [M1 A1]

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